

Agentic AI Interview Questions & Answers for Senior QA Automation Engineers

1. What is Agentic AI and how is it different from traditional AI automation?

Answer:

Traditional AI systems usually respond to a single prompt or perform one predefined task. Agentic AI systems can: • reason • plan • take decisions • execute actions • use tools • self-correct • operate across multiple steps autonomously

Example: A normal chatbot answers a question. An AI agent can: • read Jira tickets • generate test cases • execute API validations • analyze logs • create defect reports automatically

As a senior QA engineer, understanding the transition from rule-based automation to autonomous AI workflows is extremely important.

2. How can Agentic AI transform QA automation?

Answer:

Agentic AI can significantly improve: • test generation • test maintenance • defect analysis • root cause analysis • flaky test detection • self-healing automation • production monitoring

Examples: • AI generating Playwright tests from requirements • AI analyzing failed executions automatically • AI suggesting impacted regression suites • AI prioritizing high-risk areas

Future QA teams will focus more on validation strategy and less on repetitive scripting.

3. What is the role of LLMs in AI-driven testing?

Answer:

Large Language Models (LLMs) help systems understand: • natural language • business requirements • user stories • test scenarios • logs and error messages

In testing, LLMs can: • generate test cases • explain failures • convert requirements into automation • generate API payloads • create SQL queries • summarize reports

Senior engineers must understand both the power and limitations of LLMs, especially hallucination risks.

4. What are AI agents in testing?

Answer:

AI agents are autonomous systems capable of: • perception • reasoning • memory • planning • execution

In QA, an AI agent may: 1. Read a Jira story 2. Generate test cases 3. Execute tests 4. Analyze failures 5. Raise defects 6. Suggest fixes

This is very different from static automation frameworks.

5. What are the major risks of Agentic AI in testing?

Answer:

Key risks include: • hallucinated outputs • incorrect validations • security issues • biased decisions • over-automation • lack of explainability • unstable autonomous actions

Example: An AI system may incorrectly mark failed tests as passed due to misinterpretation.

Senior QA engineers must always validate AI-generated outcomes instead of blindly trusting them.

6. Explain hallucination in LLMs and how QA should validate it.

Answer:

Hallucination occurs when an LLM generates incorrect or fabricated information confidently.

Examples: • fake API responses • incorrect assertions • invalid business rules • imaginary endpoints

Validation strategies: • deterministic validations • human review • cross-verification • grounding using real data • prompt engineering

QA teams must design validation layers around AI systems.

7. How would you test an AI chatbot application?

Answer:

Testing an AI chatbot requires validating: • correctness • relevance • context retention • memory • toxicity • response latency • hallucinations • edge cases

Important areas: • prompt injection testing • multilingual validation • security testing • conversational flow testing • fallback handling

Unlike traditional systems, AI outputs are probabilistic, not deterministic.

8. What is prompt engineering and why is it important for QA?

Answer:

Prompt engineering is the process of designing prompts that produce reliable AI responses.

Good prompts improve: • response quality • consistency • structured outputs • automation reliability

Example: Instead of: "Generate tests"

Use: "Generate positive, negative, and boundary Playwright test cases for login API in JSON format."

Senior QA engineers working with AI tools must learn structured prompting.

9. How would you validate AI-generated test cases?

Answer:

Validation should include: • requirement coverage • business logic accuracy • edge cases • negative scenarios • security validations • duplicate detection

AI-generated tests should never be directly trusted in enterprise environments.

Human review remains essential.

10. Explain self-healing automation.

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<b>Answer:</b><br/><br/>
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Self-healing automation uses AI to recover from failures automatically.<br/><br/>
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Example:  
If locator changes:
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Old locator:  
#submit-btn
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New locator:  
#submit-button
```

AI may identify the updated element automatically.

Benefits: • reduced maintenance • faster recovery • improved stability

Risks: • incorrect healing • false positives

11. What are AI observability and monitoring?

Answer:

AI observability means monitoring: • model behavior • accuracy • drift • hallucinations • latency • token usage • failures

Senior QA engineers should track: • response consistency • production anomalies • unexpected outputs • prompt failures

12. What is RAG (Retrieval-Augmented Generation)?

Answer:

RAG combines: • external knowledge retrieval • LLM reasoning

Instead of relying only on model memory, the AI retrieves real-time information from: • databases • PDFs • enterprise docs • APIs

Benefits: • reduced hallucinations • better accuracy • enterprise grounding

13. How would you test AI agent workflows?

Answer:

Testing should validate: • decision-making • tool usage • workflow sequencing • memory handling • retries • fallback logic • security boundaries

Example: Validate whether an AI agent: • calls correct APIs • handles failures correctly • avoids infinite loops

14. What are autonomous workflows in QA?

Answer:

Autonomous workflows allow AI systems to execute tasks independently.

Examples: • AI-triggered regression execution • defect triaging • test impact analysis • release risk prediction

Senior QA engineers must define governance controls around these workflows.

15. How can AI help with flaky test detection?

Answer:

AI can analyze: • historical failures • timing patterns • environment instability • network issues • inconsistent assertions

AI may identify flaky patterns automatically and suggest fixes.

This reduces maintenance overhead significantly.

16. What are ethical concerns in AI testing?

Answer:

Major ethical concerns: • bias • privacy leakage • misinformation • unfair decision-making • unsafe automation

QA teams must ensure: • fairness • explainability • accountability • compliance

17. Explain AI model drift.

Answer:

Model drift occurs when AI accuracy decreases over time due to changing data patterns.

Examples: • customer behavior changes • new business scenarios • evolving language patterns

QA engineers should monitor: • prediction quality • output consistency • regression trends

18. How would you integrate AI testing into CI/CD?

Answer:

Integration includes: • AI validation pipelines • automated prompt testing • regression checks • observability dashboards • production monitoring

Important considerations: • cost optimization • token monitoring • response validation • fallback mechanisms

19. What skills should a Senior QA Engineer learn for Agentic AI?

Answer:

Important skills: • Playwright/Selenium • APIs • Python/JavaScript • prompt engineering • AI testing strategies • LLM validation • vector databases • RAG concepts • observability • AI governance

Future QA roles will require both testing and AI understanding.

20. How do you see the future of QA with Agentic AI?

Answer:

QA will evolve from: manual execution → intelligent validation engineering.

Future testers will: • supervise AI agents • validate AI decisions • design AI safety layers • manage autonomous workflows • focus on risk analysis and governance

The role of QA will become more strategic and engineering-focused rather than repetitive execution-based.
